

DB DATA MANAGER

INTRODUCTION

DB Data Manager is built using Microsoft Excel and allows data from database tables to be viewed, added and edited using Excel Worksheets. Database table schema details can also be viewed.

DB Data Manager is distributed in the hope that it will be useful for assisting with database systems administration and legacy application automation (e.g., for regression testing).

Important: All data changes are always the responsibility of the user(s) and must be checked thoroughly by the user(s), both before being applied to the database and after the databases updates have been completed. It is strongly recommended that data backups are taken before any data changes are applied.

DB Data Manager is free and open-source software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version.

DB Data Manager is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY, without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

The project files with source code can be found at <https://github.com/christyler80/DBExcelDataManager>.

DB Data Manager requires Microsoft Excel 2016 or later and supports SQL Server, SQLite, ODBC (32bit) and Microsoft Access.

Basic testing has been undertaken using the Northwind and Pubs databases running on [SQL Server](#), [SQLite](#), [MySQL/MariaDB](#) and [Microsoft Access](#).

Where possible, please use Northwind or Pubs database examples to report any errors and/or feature requests.

Important: At this time:

- 32-bit or 64-bit versions of Microsoft Windows 10 operating system or later are supported.
- Excel 2019 and later versions are supported. Earlier Excel versions may work but have not been tested.
- Other operating systems (e.g., Apple, Linux) are not currently supported.
- Most text and numeric field data types are supported.
- Binary data and other non-text or non-numeric field types are not supported.
- Data must be entered with correct formatting and numeric scale/precision (e.g., number of decimal places).
- DB Data Manager should work on Windows 7, but this has not been tested.
- Other operating systems (e.g., Apple, Linux) are not supported.

FUTURE VERSIONS

Planned future enhancements for DB Data Manager include support for other DB servers (e.g., Oracle, Postgres), deeper database schema integration and equivalent VB.NET and C# programs.

Please provide feedback via GitHub and/or by contacting me using the details below.

Chris Tyler, September 2025

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GETTING STARTED

- Before using DB Data Manager, ensure that all required database connectivity software (e.g., 32-bit ODBC drivers for the target databases and servers) is installed and connections to the target database(s) are working correctly.
- Using an empty worksheet, confirm that Excel will connect to the target database(s), using Excel's own built in data tools.
- Download DB_DataManager.xlsm workbook from https://github.com/christyler80/DB_DataManager copy into a folder and rename if desired.
- Open the Excel workbook and ensure that Excel macros are enabled.
- The examples in this document show data from a Microsoft Access Northwind database running on the local machine.

Supplier IDs	Product Code	Product Name	Description	Standard Cost	List Price	Reorder Level	Target Level	Quantity Per Unit	Discontinued	Minimum Reorder Quantity	Category	Attachments
1	1	Northwind Traders Chap		£13.50	£18.00	10	40	10 boxes x 20 bags	FALSE	10	Beverages	
2	2	Northwind Traders Syrup		£7.50	£10.00	25	100	12 - 550 ml bottles	FALSE	25	Condiments	
3	3	Northwind Traders Cajun Seasoning		£16.50	£22.00	10	40	48 - 6 oz jars	FALSE	10	Condiments	
4	4	Northwind Traders Olive Oil		£16.01	£21.35	10	40	36 boxes	FALSE	10	Oil	
5	5	Northwind Traders Boysenberry Spread		£18.75	£25.00	25	100	12 - 8 oz jars	FALSE	25	Jams, Preserves	
6	6	Northwind Traders Dried Pears		£22.50	£30.00	10	40	12 - 1 lb plgs.	FALSE	10	Dried Fruit & Nuts	
7	7	Northwind Traders Walnuts		£17.44	£23.25	10	40	12 - 12 oz jars	FALSE	10	Dried Fruit & Nuts	
8	8	Northwind Traders Fruit Cocktail		£29.25	£39.00	10	40	15.25 OZ	FALSE	10	Canned Fruit & Vegetables	
9	9	Northwind Traders Chocolate Biscuits Mix		£6.90	£9.20	5	20	10 boxes x 12 pieces	FALSE	5	Baked Goods & Mixes	
10	10	Northwind Traders Marmalade		£60.75	£81.00	10	40	30 gift boxes	FALSE	10	Jams, Preserves	
11	11	Northwind Traders Scopes		£7.50	£10.00	5	20	24 plgs. x 4 pieces	FALSE	5	Baked Goods & Mixes	
12	12	Northwind Traders Beer		£10.50	£14.00	15	60	24 - 12 oz bottles	FALSE	15	Beverages	
13	13	Northwind Traders Crab Meat		£13.80	£18.40	30	120	24 - 4 oz tins	FALSE	30	Canned Meat	
14	14	Northwind Traders Clam Chowder		£7.24	£9.65	10	40	12 - 12 oz cans	FALSE	10	Soups	
15	15	Northwind Traders Coffee		£34.50	£46.00	25	100	16 - 500 g tins	FALSE	25	Beverages	
16	16	Northwind Traders Chocolate		£9.56	£12.75	25	100	10 plgs	FALSE	25	Candy	
17	17	Northwind Traders Dried Apples		£39.75	£53.00	10	40	50 - 300 g plgs.	FALSE	10	Dried Fruit & Nuts	
18	18	Northwind Traders Long Grain Rice		£5.25	£7.00	25	100	15 - 2 kg boxes	FALSE	25	Grains	
19	19	Northwind Traders Gnocchi		£28.50	£38.00	30	120	24 - 250 g plgs.	FALSE	30	Pasta	
20	20	Northwind Traders Ravioli		£14.63	£19.50	20	80	24 - 250 g plgs.	FALSE	20	Pasta	
21	21	Northwind Traders Hot Pepper Sauce		£15.79	£21.05	10	40	32 - 8 oz bottles	FALSE	10	Sauces	
22	22	Northwind Traders Tomato Sauce		£12.75	£17.00	20	80	24 - 8 oz jars	FALSE	20	Sauces	
23	23	Northwind Traders Mozzarella		£26.10	£34.80	10	40	24 - 200 g plgs.	FALSE	10	Dairy Products	
24	24	Northwind Traders Almonds		£7.50	£10.00	5	20	5 kg pkg	FALSE	5	Dried Fruit & Nuts	
25	25	Northwind Traders Mustard		£9.75	£13.00	15	60	12 boxes	FALSE	15	Condiments	
26	26	Northwind Traders Dried Plum		£3.00	£3.60	60	75	1 lb bag	FALSE	25	Dried Fruit & Nuts	
27	27	Northwind Traders Green Tea		£2.00	£2.59	100	125	20 bags per box	FALSE	25	Beverages	
28	28	Northwind Traders Granola		£2.00	£4.00	20	100		FALSE		Cereal	
29	29	Northwind Traders Potato Chips		£0.50	£1.80	30	200		FALSE		Chips, Snacks	
30	30	Northwind Traders Brownie Mix		£9.00	£12.49	10	20	3 boxes	FALSE	5	Baked Goods & Mixes	
31	31	Northwind Traders Cake Mix		£10.50	£15.99	10	20	4 boxes	FALSE	5	Baked Goods & Mixes	
32	32	Northwind Traders Tea		£2.00	£4.00	20	50	100 count per box	FALSE		Beverages	
33	33	Northwind Traders Pears		£1.00	£1.30	10	40	15.25 OZ	FALSE		Canned Fruit & Vegetables	
34	34	Northwind Traders Peaches		£1.00	£1.50	10	40	15.25 OZ	FALSE		Canned Fruit & Vegetables	
35	35	Northwind Traders Pineapple		£1.00	£1.80	10	40	15.25 OZ	FALSE		Canned Fruit & Vegetables	
36	36	Northwind Traders Cherry Pie Filling		£1.00	£2.00	10	40	15.25 OZ	FALSE		Canned Fruit & Vegetables	
37	37	Northwind Traders Green Beans		£1.00	£1.20	10	40	14.5 OZ	FALSE		Canned Fruit & Vegetables	
38	38	Northwind Traders Corn		£1.00	£1.20	10	40	14.5 OZ	FALSE		Canned Fruit & Vegetables	
39	39	Northwind Traders Peas		£1.00	£1.50	10	40	14.5 OZ	FALSE		Canned Fruit & Vegetables	
40	40	Northwind Traders Tuna Fish		£0.50	£2.00	30	50	5 oz	FALSE		Canned Meat	
41	41	Northwind Traders Smoked Salmon		£2.00	£4.00	30	50	5 oz	FALSE		Canned Meat	
42	42	Northwind Traders Hot Cereal		£3.00	£5.00	50	200		FALSE		Cereal	
43	43	Northwind Traders Vegetable Soup		£1.00	£1.89	100	200		FALSE		Soups	
44	44	Northwind Traders Chicken Soup		£1.00	£1.95	100	200		FALSE		Soups	

- Database connection details are shown in row 1.
- Table information including field names and data type details are shown in rows 2 to 5.
- This information in rows 1 to 5 is read-only and should not be changed or overwritten manually.
- Table data in row 6 and below can be changed using standard Excel functionality.
- Editing numeric and text data types is supported.
- Binary data (e.g., pictures) and other non-text or non-numeric data types are not currently supported.
- Primary, Unique and Foreign keys and unsupported data types are highlighted in row 4. Data in these columns is read only for existing records.
- Always use the 'New Sheets' button to create new DB Data Manager worksheets.

SELECT DATA SOURCE

- Press the 'Data Source' button and the Select Data Source form should open. Connection details will be populated automatically when worksheet data has previous been loaded using DB Data Manager.

DB Data Admin - Select Data Source

Driver Types: SQL Server

Data Sources: (local)

Login ID:

Password:

Open Connection

Connection String: DRIVER={SQL Server Native Client 11.0};SERVER=(local);Trusted_Connection=Yes;Database=Northwind

Databases:

- Northwind
- Pubs
- ScratchpadDB

Tables:

- Categories
- CustomerCustomerDemo
- CustomerDemographics
- Customers
- Employees
- EmployeeTerritories
- Order Details
- Orders
- Products
- Region
- Shippers
- Suppliers

Get Schema Load Data Run Sql Cancel

- Select the required Driver Type and Data Source.
- Enter Login ID and Password, else leave blank if Windows Authentication is to be used.
- Press the 'Open Connection' button, to list available databases and tables. Select the required database and table.
- Press 'Load Data' button to load data from the selected table. When this exceeds 250 records, a form will appear allowing SQL to be used to filter the data being loaded into Excel. It is recommended to keep the number of records being loaded for editing to less than 250 in any single session.

DB Data Admin - Select Data Source

Driver Types: Microsoft Access

Data Sources: C:\Development\DB Data Manager\North

Login ID:

Password:

Open Access Database

Open Connection

Connection String: Provider=Microsoft.Ace.Oledb.12.0;Data Source=C:\Development\DB Data Manager\Northwind.accdb;

Databases:

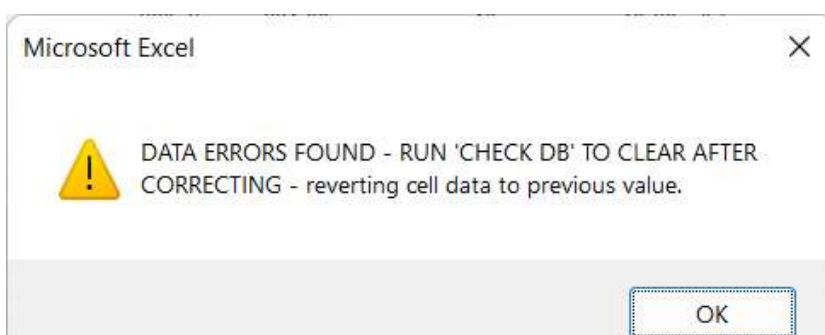
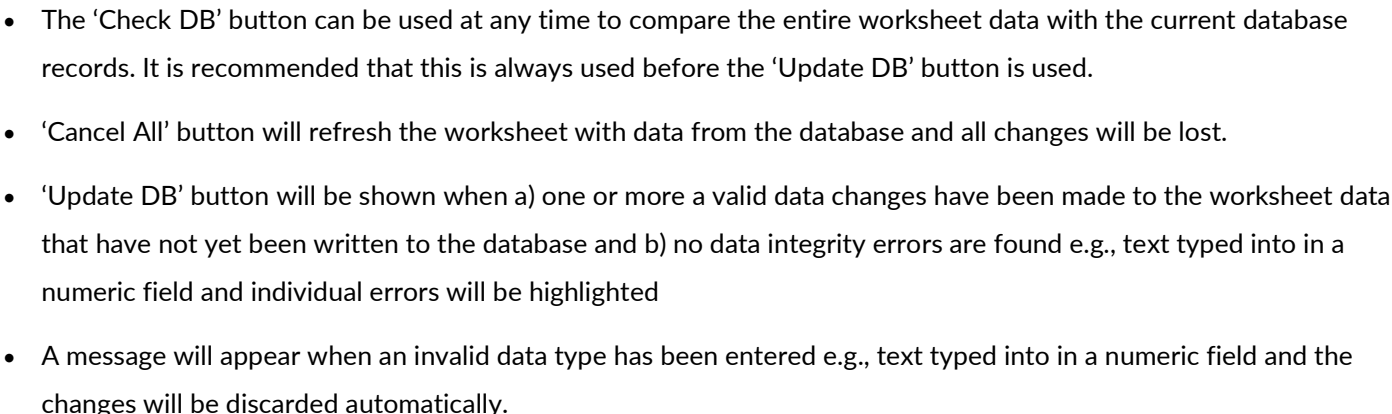
- Northwind.accdb;

Tables:

- Customers
- Employee Privileges
- Employees
- Inventory Transaction Types
- Inventory Transactions
- Invoices
- Order Details
- Order Details Status
- Orders
- Orders Status
- Orders Tax Status
- Privileges

Get Schema Load Data Run Sql Cancel

- Database and Table information in rows 1 to 5 is read-only and should not be changed or overwritten manually.
- Table data in row 6 and below can be changed using standard Excel functionality.
- Valid worksheet data changes will be highlighted automatically after each change when the 'Checks On' button displayed. When 'Checks Off' is displayed, data changes are highlighted without any validation checks.



DATA EDITING continued,

- Pressing the 'Checks On/Off' button can be used to turn automatic data validation checking On or Off.
- Data changes made to the worksheet with 'Checks Off' selected will be highlighted in yellow as 'Unverified'.

The screenshot shows the Microsoft Excel interface with the 'Checks Off' button highlighted in the ribbon. The worksheet displays a table of data with columns for Supplier IDs, Product Code, Product Name, Standard Cost, List Price, Reorder Level, Target Level, Quantity Per Unit, Discontinued, Minimum Reorder Quantity, Category, and Attachments. The 'Checks Off' button is located in the ribbon, and the 'Data change checking turned Off' message is visible in the status bar. The worksheet shows data for various products, including Northwind Traders items, with some cells highlighted in red to indicate data validation errors.

- The 'Check DB' button can be used at any time to compare the entire worksheet data with the current database records. Invalid data types such as text in a numeric field, will be highlighted in 'Red'.
- 'Cancel All' button will refresh the worksheet with data from the database and all changes will be lost.

The screenshot shows the Microsoft Excel interface with the 'Check DB' button highlighted in the ribbon. The worksheet displays a table of data with columns for Supplier IDs, Product Code, Product Name, Standard Cost, List Price, Reorder Level, Target Level, Quantity Per Unit, Discontinued, Minimum Reorder Quantity, Category, and Attachments. The 'Check DB' button is located in the ribbon, and the 'Data change checking turned Off' message is visible in the status bar. The worksheet shows data for various products, including Northwind Traders items, with some cells highlighted in red to indicate data validation errors.

SAVING DATA CHANGES TO THE DATABASE

- 'Update DB' button will be visible when a) one or more valid data changes have been made to the worksheet data that have not yet been written to the database and b) no data integrity errors are found e.g., text typed into in a numeric field and individual errors will be highlighted.

The screenshot shows the DB Manager application interface. The 'Update DB' button is highlighted in red, indicating that data changes can be saved to the database. The application window displays a list of products and their details, including product names, descriptions, and various attributes like standard cost, list price, and reorder level. The 'Update DB' button is located in the top right corner of the application window.

- Pressing the 'Update DB' button will update existing database records with any data changes in the worksheet.
- DB updates will be cancelled automatically if an error is encountered. Successful updates will be highlighted in 'Blue'

The screenshot shows the DB Manager application interface. The 'Update DB' button is highlighted in blue, indicating that data changes have been successfully saved to the database. The application window displays a list of products and their details, including product names, descriptions, and various attributes like standard cost, list price, and reorder level. The 'Update DB' button is located in the top right corner of the application window.

TABLE SCHEMA INFORMATION

- ‘Get Schema’ button will list [Microsoft ActiveX Data Objects \(ADO\)](#) schema information for the current data table including [data types](#) and index information.

AutoSave DB_DataManager.xlsx Search Chris Tyler													
File Home Insert Page Layout Formulas Data Review View Developer Help Easy Document Creator													
D16 Products													
Data Source: Microsoft Access Provider=Microsoft.Ace.OleDB.12.0;Data Source=C:\Development\DB Data Manager\Northwind.mdb													
Data Table: Products													
Data Source Get Schema Get Data													
New Sheet													
adSchemaTables TABLE_CATALOG TABLE_SCHEMA TABLE_NAME TABLE_TYPE TABLE_GUID DESCRIPTION TABLE_PROPID DATE_CREATED DATE_MODIFIED													
Show Other Tables													
adSchemaPrimaryKeys TABLE_CATALOG TABLE_SCHEMA TABLE_NAME COLUMN_NAME COLUMN_GUID COLUMN_PROPID ORDINAL PK_NAME													
1 PrimaryKey													
adSchemaIndexes TABLE_CATALOG TABLE_SCHEMA TABLE_NAME INDEX_CATALOG INDEX_SCHEMA INDEX_NAME PRIMARY_KEY UNIQUE CLUSTERED TYPE FILL_FACTOR													
Attachments_4962043875874 TRUE FALSE 1 100													
PrimaryKey TRUE TRUE FALSE 1 100													
Supplier IDs_CPF83008C7136 FALSE TRUE FALSE 1 100													
Product Code FALSE FALSE FALSE 1 100													
adSchemaForeignKeys PK_TABLE_CATALOG PK_TABLE_SCHEMA PK_TABLE_NAME PK_COLUMN_NAME PK_COLUMN_GUID PK_COLUMN_PROPID FK_TABLE_CATALOG FK_TABLE_SCHEMA FK_TABLE_NAME FK_COLUMN_NAME FK_COLUMN_GUID													
Primary Table Product ID Inventory Transactions Product ID													
Foreign Table Order Details Order Details Product ID													
adSchemaColumns TABLE_CATALOG TABLE_SCHEMA TABLE_NAME COLUMN_NAME COLUMN_GUID COLUMN_PROPID ORDINAL_POSITION COLUMN_HASDEFAULT COLUMN_DEFAULT COLUMN_FLAGS IS_NULLABLE													
Products Attachments 14 FALSE 234 TRUE													
Products Category 13 FALSE 195 TRUE													
Products Description 5 FALSE 234 TRUE													
Products Discontinued 11 TRUE No 90 FALSE													
Products List Price 2 FALSE 90 FALSE													
Products Minimum Reorder Quantity 7 TRUE 0 122 TRUE													
Products Product Code 3 FALSE 106 TRUE													
Products Product Name 4 FALSE 106 TRUE													
Products Quantity Per Unit 10 FALSE 106 TRUE													
Products Reorder Level 8 FALSE 122 TRUE													
Products Standard Cost 6 TRUE 122 TRUE													
Products Supplier IDs 1 FALSE 234 TRUE													
Products Target Level 9 FALSE 122 TRUE													
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Products Minimum Reorder Quantity FALSE 1033													
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Products Target Level FALSE 1033													
DB Errors Log DB Updates Log Northwind MS Access ADO Data Types													
Ready Calculate Accessibility: Investigate													

- Pressing the ‘Get Data’ button will retrieve data records for the current table (e.g., ‘Products’ in the example below).

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File Home Insert Page Layout Formulas Data Review View Developer Help Easy Document Creator													
A5 Supplier IDs													
Data Source: Microsoft Access Provider=Microsoft.Ace.OleDB.12.0;Data Source=C:\Development\DB Data Manager\Northwind.mdb													
Data Table: Products 45 RECORDS LOADED													
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1 NWITB-1 Northwest Traders Chili													
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3 NWITC-4 Northwest Traders Cajun Seasoning													
4 NWITC-5 Northwest Traders Olive Oil													
5 NWITP-6 Northwest Traders Boysenberry Spread													
6 NWITP-7 Northwest Traders Dried Pears													
7 NWITP-8 Northwest Traders Curry Sauce													
8 NWITP-9 Northwest Traders Walnuts													
9 NWITP-10 Northwest Traders Fruit Cocktail													
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